

Project Name: Unbrokey

Team Members: Mariana Piñero Cavazos, Camila Rodriguez Rodriguez, Camila Valladares De la Torre

Course Name: Specification of Products and Services

Session Number: Session 3

Date: 21/05/2026



This material and **CMF (Color, Material, Finish)** proposal for **Unbrokey** is based on the project's core philosophy of **extreme resistance and real accessibility**. It translates the technical requirements of a high-performance modular system into a tangible product direction.

Material Strategy

The strategy is defined by the "**Unbroken**" philosophy: prioritizing **quality and reliability** over price or aesthetics alone. The goal is to create a "Performance-Tool" aesthetic that fosters **trust and long-term habit building**. This involves a hybrid material approach—combining structural polymers, industrial fasteners, and high-strength magnets—to ensure the system remains **tough but light**.



Quality and Reliability Over Price

A central tenet of the strategy is prioritizing **high quality and reliability** as the top decision drivers. User research indicates that target users are willing to invest more in a product that delivers **long-term value and durability**. The goal is to provide users with **reliability, security, and peace of mind**, ensuring that the system can withstand the rigors of an active daily routine without failure.

The "Tough but Light" Objective

The strategy aims to balance **extreme resistance** with **portability**. Materials are selected and integrated to ensure the system remains **lightweight** for easy transportation in a backpack, yet **tough enough** to serve as a functional external workstation in high-impact environments like gyms.

Layered Structural Integrity

The strategy utilizes a **hybrid structural approach** to maximize functionality. By using a layered "sandwich" design for the main board, the system combines different functional properties—such as structural rigidity, modular attachment, and surface mounting capabilities—into a single, compact form factor.

Frictionless and Secure Interaction

The material strategy focuses on **improving the user experience** through tactile and functional performance. This includes selecting components that allow for:

- **One-hand operation** for quick attachment and detachment.
- **High-friction interfaces** to ensure carried items remain secure during movement.
- **Safety-first design**, utilizing rounded edges and smooth finishes to prevent injury or snagging in active environments.

Localized Quality Control

A strategic decision has been made to focus on **local manufacturing and assembly**. this allows the team to maintain a **close eye on quality** throughout the production process, ensuring that the final output aligns with the brand's premium standards and "built to last" commitment.

Sustainability Through Longevity

The strategy views **sustainability as a core value**. Instead of relying on low-cost, disposable materials, the focus is on **long-lasting components** that reduce the need for frequent replacement. This approach supports a **long product life cycle** and aligns with the emerging consumer mindset of "resilient self-care" and value-based consumption

Main Parts and Possible Materials

The **Unbrokey** system is composed of several key structural and functional components designed to work together as a durable, modular ecosystem. The architecture of the product relies on a **layered "sandwich" construction** for the main board and high-performance materials for its interchangeable modules.

The Base Board (Layered Structure)

The foundation of the system is a horizontal board consisting of three distinct layers:

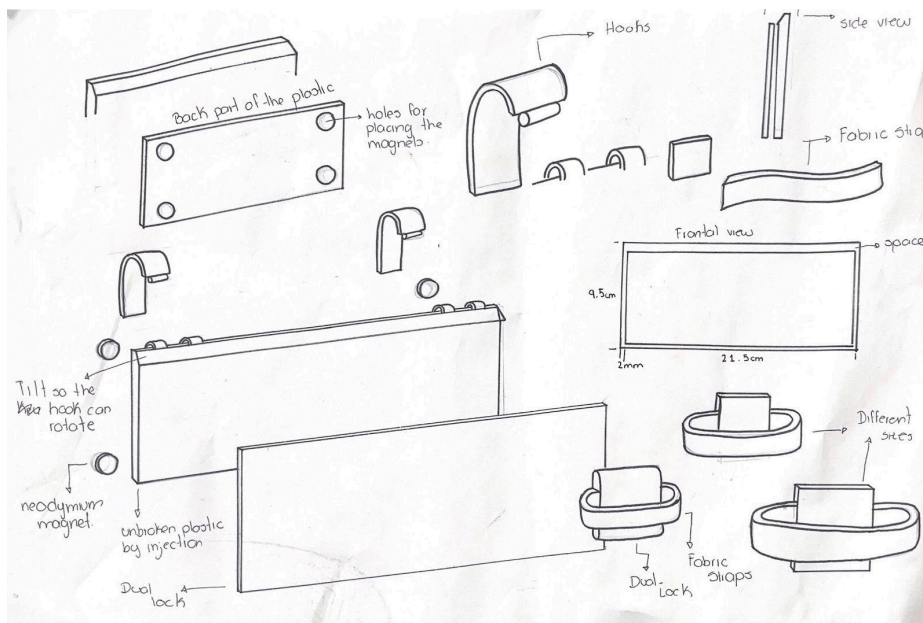
- **Dual-Lock Velcro (Front Face):** This layer acts as the primary attachment interface for all modules. It is described as an **improved version of standard Velcro** with high cycle-life and industrial strength to ensure items stay secure during movement.
- **Unbroken Plastic Plate (Structural Core):** The central structural component is made from "**Unbroken**" plastic, a polymer designed to be **injection-molded** for high precision. This core is engineered to be **tough but light**, providing the necessary rigidity without adding unnecessary weight to the user's bag.
- **Metal Magnetic Plate (Back Face):** The rear layer consists of a **magnetic metal sheet**, which allows the entire board to be mounted onto external metallic surfaces, such as gym racks, lockers, or refrigerators.

Mounting and Attachment Components

- **Neodymium (NdFeb) Magnets:** The system integrates **four circular high-strength neodymium magnets**. These are embedded into the board or keychain to facilitate a reliable "**snap-on**" attachment to gym equipment, allowing for hands-free access.
- **Hooks and Dock Latches:** To attach the system to backpacks or bags, Unbrokey uses **metal rotating hooks** and specialized dock latches. The use of metal ensures these points of failure in traditional organizers remain **extremely resistant** under load.
- **Hexagonal Structural Frame:** An iconic part of the design is the **hexagonal central frame**, which serves as a main structural anchor and provides multiple points for clipping accessories while reinforcing the brand's geometric aesthetic.

Interchangeable Modules and Retention Parts

- **Item Retention Straps:** These are used to secure items like AirPods, wallets, or chargers to the board. They are crafted from **nylon elastic webbing** featuring integrated **silicone/grip lines** on the interior to increase friction and prevent items from slipping out.
- **Interchangeable Modules:** These include clips, holders, and **hard secure boxes** made from a combination of **plastic and fabric**. These modules are designed to be "modular and adaptable," allowing users to add or remove them based on their daily routine.
- **Velcro Item Straps:** Specific square attachments with the opposite side of the Dual-Lock to interface directly with the main board



3. Material Properties Needed

The **Unbrokey** system is a modular carrying solution designed under the philosophy of **extreme resistance and real accessibility**. It is engineered to organize essential items—such as keys, wallets, and headphones—while reducing the friction and stress caused by disorganized bags during daily transitions.

- **Resistance and Durability:** The system must withstand impacts, drops, and constant use in gym environments without breaking or deforming. To achieve this, it utilizes a structural core made of **Unbroken plastic** produced via **injection molding** for high precision, combined with a front face of **heavy-duty Dual Lock Velcro**.
- **Magnetic Strength:** The design integrates **four circular high-strength neodymium (NdFeb) magnets**. These allow for a firm, "snap-on" attachment to metal surfaces like gym racks or refrigerators, keeping items visible and hands-free.
- **Lightweight and Portability:** A critical priority is that the board remains **lightweight** to avoid adding physical strain or "logistical complexity" for the user. The objective is a body that is specifically **"tough but light"**.
- **Tactile Grip and Friction:** The item retention straps are crafted from **nylon elastic webbing** featuring integrated **silicone/grip lines** on the interior. This increases friction to prevent smooth objects, like headphone cases, from slipping out during movement.
- **Safety and Ergonomics:** The design prioritizes **rounded edges** and smooth surfaces to prevent snagging on clothing or causing injury during active routines. It is also optimized for **one-hand operation**, allowing users to attach or detach the board and modules effortlessly.
- **Longevity and Sustainability:** Materials are selected for a **long life cycle**, reducing the need for frequent replacement. This aligns with the brand's commitment to sustainability and the emerging consumer mindset of "resilient self-care".

In summary, these properties transform Unbrokey from a simple organizer into a **performance-enhancing tool** that provides the user with reliability, security, and peace of mind

Possible Production Processes

Unbrokey system is centered on achieving a balance between extreme durability and lightweight portability through precise manufacturing and high-quality assembly.

Injection Molding

This is the primary production process used for the **main structural body** of the horizontal board.

- **Purpose:** It is strategically selected to ensure the "Unbroken plastic" core remains **"tough but light,"** providing the necessary impact resistance for gym and urban environments.
- **Precision:** Molding allows for high-precision features, such as the **top tilt** required for hook rotation and specific **slots or holes** for integrating magnets.

Component Integration and Bonding

The product is a multi-material system that requires the integration of several distinct layers and parts:

- **Layering:** The process involves bonding the **Dual Lock Velcro** to the front face and attaching the **metal magnetic plate** to the rear of the plastic core.
- **Magnetic Embedding:** Four circular **neodymium (NdFeb) magnets** are integrated into the keychain or base board to ensure secure "snap-on" functionality.
- **Manual/Mechanical Assembly:** The assembly of interchangeable modules—such as clips, metal rotating hooks, and **nylon elastic straps** with silicone grip lines—is required to complete the ecosystem.

Local Manufacturing and Assembly

A key part of the production strategy is to manufacture and assemble the product **locally**.

- **Quality Control:** By keeping production local, the team maintains a **close eye on quality** and ensures that every unit meets the "built to last" standards of the brand's philosophy.
- **Reliability:** Local assembly ensures that high-strength components (magnets, Velcro, and metal hooks) are securely bonded and tested for **reliability and safety** before reaching the user.

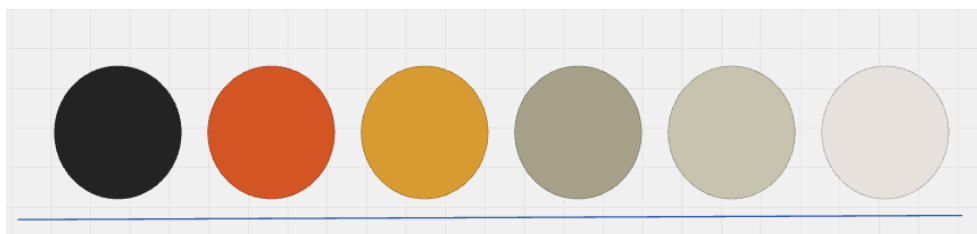
Material Fabrication (Modules)

Secondary processes are necessary for the accessory modules:

- **Textile Processing:** The retention straps are made from **nylon elastic webbing** that includes specialized **silicone/grip line patterns** to increase friction.
- **Metal Component Sourcing:** Structural elements like the rotating hooks and dock latches are made of **metal**, requiring standard metal forming or casting processes

CMF Palette Direction

- **Color Palette:** A professional, high-contrast mix of **Black, Orange, Grey, and Beige**.



- **Finishes:**
 - **Plastic:** Textured matte finish (anti-scratch and ergonomic grip).
 - **Metal:** Brushed or coated to prevent corrosion from sweat.
 - **Fabric:** Tactical-style matte nylon webbing .

Material and CMF Summary Table

Part / Component	Proposed Material	Key Properties	Process	Finish / Color	Risk / Alternative
Main Base Board	Unbroken Plastic	Tough, light, injection-ready	Injection Molding	Matte Black or Grey	To be defined (Polymer grade)
Front Interface	Heavy-duty Dual Lock Velcro	High cycle life; improved interlocking strength	Adhesive bonding to core	Textured/black	Adhesive failure in high heat / Needs testing
Mounting Magnets	Neodymium (NdFeb)	High-strength retention	Integration into board	Circular / Hidden	Interference with tech items
Item Straps	Nylon + Silicone Grip	Elasticity and friction	Weaving/Assembly	Vibrant Orange	Wear over time / Elasticity loss
Attachment Face	Dual Lock Velcro	High cycle life, heavy-duty	Adhesive bonding	Black Textured	Adhesive failure in heat
Hooks / Latch	Metal	Structural rigidity	Forming	Dark Grey or Black	Needs testing for one-hand use

Technical Report Communication

In the technical manufacturer report, materials should be explained through **reliability and precision**. The focus must be on **compliance with safety standards**, such as secure magnetic integration and **rounded edges** to prevent injury. Technical success will be measured by **load resistance and attachment cycles** (e.g., how many times the Dual Lock can be used before losing grip).

Visual / Commercial Communication

The **Visual and Commercial Communication** strategy for **Unbrokey** focuses on positioning the product as a premium "performance-enhancing tool" rather than a simple organizer. This is achieved through a multi-channel approach that emphasizes **reliability, high quality, and real-life functionality**.

1. Core Commercial Deliverable: Dedicated E-commerce Platform

The primary commercial hub will be a **dedicated e-commerce website** designed to reflect the brand's commitment to quality.

- **Central Sales Hub:** It will serve as a platform to purchase **Starter Kits**, bundle packs, and individual interchangeable modules for full customization.

- **Trust Building:** To foster consumer confidence, the site will feature **strong visual storytelling** and detailed technical specifications regarding the **"Unbroken" plastic, NdFeb magnets, and Dual Lock Velcro.**
- **User Support:** The platform will integrate **user tutorials** and a **return system for lost items** to ensure a seamless transition into a modular lifestyle.

2. Visual Content and Social Media Strategy

Visual communication relies heavily on **short-form digital content** (TikTok, Instagram Reels, and YouTube) to bridge the gap between technical innovation and daily usage.

- **"Before vs. After" Storytelling:** Content will highlight the transition from a messy, stressful bag to an organized, reachable workstation.
- **Real-Life Routines:** Videos will demonstrate how the product integrates into daily routines across **gym, work, and school environments.**
- **Technical Highlighting:** Visuals will utilize **high-fidelity renders, 3D screenshots, and prototype photos** to document real-world user interaction and product durability.

3. Physical Presence and Experiential Marketing

To complement digital efforts, Unbrokey utilizes **BTL (Below the Line) activations** and strategic partnerships.

- **BTL Activations:** Physical stands or displays in retail spaces (like gyms or supermarkets) allow potential customers to **touch, feel, and try** the modular system in real time.
- **Cross-Promotion:** The brand will partner with **gyms and fitness centers** to offer the product as part of membership deals or promotional gifts, increasing perceived value.
- **Key Partnerships:** Collaboration with **content creators and marketing agencies** will be used to validate the product through community-driven feedback.

4. Strategic Positioning

The commercial message focuses on **changing the organizational paradigm.** Unlike traditional organizers that are often fixed or bulky, Unbrokey is marketed as an **accessible, adaptable, and multi-context platform.** The communication emphasizes that the product helps users stay focused on their day by providing **frictionless access** to essentials without the stress of losing them